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FedSubMerge: Subspace Merging for Replay-Free Federated Continual Medical Image Classification

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***Abstract*— Federated learning enables medical institutions to collaboratively train diagnostic models without sharing raw patient data. However, most medical image studies assume stationary tasks and data distributions. This assumption does not reflect real-world clinical deployments, where diagnostic demands and data characteristics evolve over time. We propose FedSubMerge, a replay-free federated continual learning framework for medical image classification. FedSubMerge addresses global catastrophic forgetting during sequential training across heterogeneous clients. In this setting, local updates can interfere with knowledge acquired from tasks observed at other institutions. Each institution summarizes its historical knowledge with a compact principal gradient subspace. The server then merges these subspaces into a federated subspace that guides later local updates away from interfering directions. However, merging all client subspaces can over-constrain learning under client heterogeneity and reduce plasticity. We therefore propose FedSubMerge-AD. It uses layer-wise subspace distances to merge only related clients, which reduces excessive constraints and improves model plasticity. We evaluated the proposed framework on four medical-image benchmarks covering label skew, quantity skew, and real-world multi-center feature heterogeneity. Across these settings, our methods demonstrated competitive average accuracy and backward transfer with modest additional communication overhead. The source code will be made publicly available.**

***Index Terms*— Medical image classification, Federated continual learning, Catastrophic forgetting.**

I. Introduction

FEDERATED learning (FL) allows hospitals to train a shared model without sharing raw patient data across institutions [1], [2]. Most FL studies assume a fixed task and a stable data distribution. Real-world clinical practice, however, is not stationary. New diagnostic targets, acquisition protocols, devices, and patient populations can change the task sequence and local data distributions over time [3].

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A useful cross-site model must learn these changes without losing knowledge from earlier tasks. This setting is known as federated continual learning (FCL) [4], [5]. We study replay-free FCL, where only the current-task data are available during training. This constraint is important for medical applications because storing, transferring, or reusing earlier patient images may conflict with local consent requirements or data storage and use policies.

In FCL, especially under non-IID data, forgetting is not confined to individual hospitals. Here, *global catastrophic forgetting* refers to the decline in the server-maintained global model's performance on previously learned tasks across the federation as new tasks are learned [5]. Unlike local forgetting, which concerns the task history of a single institution, global forgetting reflects the loss of prior knowledge accumulated across institutions. Because each hospital updates the model using only its current local data, statistical heterogeneity can cause local updates to drift from the global objective. Aggregating these heterogeneous updates can further amplify forgetting.

Existing replay-free methods can mitigate forgetting through regularization or gradient projection. Gradient Projection Memory (GPM) [6] stores the main gradient directions of earlier tasks and projects new gradients away from them. However, GPM was designed for a single learner. If each client uses GPM independently, its stored gradient subspace covers only its own task history. It cannot include tasks observed only at other sites. This creates the first challenge addressed in this work: extending gradient-subspace projection from local task histories to task knowledge distributed across institutions.

Even after sharing information across clients, a second challenge remains. Medical clients can differ in label prevalence, scanners, acquisition conditions, and task histories. Consequently, some directions associated with previous tasks may be irrelevant to a particular client or layer. Keeping all such directions can shrink the feasible update space. It can therefore over-constrain the incoming task and reduce model plasticity. In this work, *plasticity* denotes the model's ability to learn an incoming task by retaining effective parameter-update directions. A practical server-side mechanism should preserve useful information from other institutions while excluding directions that unnecessarily restrict a target client's updates.

We propose Federated Subspace Merging (FedSubMerge), a replay-free FCL method that combines client gradient subspaces at the server. After each task, every participating client

updates a low-rank *principal gradient subspace* (PGS) from its gradient history. It then uploads the basis and singular values. The server merges these PGSs, and FedSubMerge returns one shared PGS to all clients. The shared PGS helps preserve task knowledge distributed across institutions. **FedSubMerge-AD** addresses the second challenge. It measures layer-wise subspace distances and selects related peers separately at each layer. The server then returns a personalized PGS that excludes less relevant directions associated with previous tasks. During the next task, each client projects gradients onto the orthogonal complement of the returned PGS. This projection reduces interference with earlier tasks while preserving plasticity for the incoming task. Clients send low-rank PGSs alongside standard model updates. No raw images or replay data are transmitted.

Our main contributions are as follows:

- We propose FedSubMerge, a replay-free framework for mitigating global catastrophic forgetting in federated continual medical image classification. The server merges client PGSs into a shared protective subspace, extending protection from local task histories to knowledge distributed across institutions. Each client then projects its local gradients onto the orthogonal complement of the returned subspace, preserving prior knowledge without storing or replaying previous patient images.
- Building on FedSubMerge, we propose FedSubMerge-AD, which performs layer-wise peer selection according to subspace distance and constructs client-specific PGSs. It reduces excessive constraints and preserves model plasticity under client heterogeneity. This flexibility improves the model's ability to learn newly encountered diseases.
- We evaluate the proposed framework on four medical imaging benchmarks across seven federated settings. These settings cover distribution-based label skew, quantity-based label skew, and real-world multi-center feature skew. The proposed methods achieved competitive performance across these settings, while introducing modest additional communication overhead.

II. Related Work

FedSubMerge lies at the intersection of federated medical image analysis, continual learning, and federated continual learning. We therefore review these three lines of work.

A. Federated Medical Image Analysis

Federated learning (FL) allows medical institutions to train a shared model without centralizing raw data [1], [2]. General FL methods address client differences through proximal regularization [7] or control variates [8]. Other approaches personalize model components [9]–[11]. Medical FL methods also address differences between sites. FedBN keeps batch-normalization statistics local [12], HarmoFL aligns frequency-domain information across sites [13], and FedDG studies generalization to unseen sites [14]. Most of these methods assume a fixed task and a stable label space. FedSubMerge instead studies a replay-free continual setting in which new diagnostic tasks arrive over time and earlier images are no longer available.

B. Continual Learning and Gradient-Subspace Methods

Continual learning (CL) aims to learn a sequence of tasks while preserving knowledge from earlier tasks. Replay-based methods store or generate samples from previous tasks [15]–[18]. However, replay may be unsuitable in multi-institutional medical settings. Local rules can limit the storage or reuse of patient images. Regularization-based methods penalize changes to important parameters or predictions [19]–[22]. They do not require earlier images, but may become less effective over long task sequences or large data shifts. Parameter-isolation methods reserve separate parameters or modules for different tasks [23]–[26], but their stored state grows as more tasks are learned.

Gradient-subspace methods provide a compact replay-free approach. They store important gradient directions from earlier tasks and project new gradients away from those directions. OGD uses gradients from earlier samples [27], while GPM and its variants estimate gradient subspaces from layer representations [6], [28], [29]. These methods were designed for a single learner that observes the full task sequence. In FL, each client observes only its local task history. FedSubMerge extends this approach to distributed task histories. Clients update their PGSs locally, and the server merges them. It then returns shared or personalized PGSs for later local training.

C. Federated Continual Learning

Federated continual learning (FCL) combines client heterogeneity with sequential task arrival. FedWeIT uses task-specific parameter decomposition [4], while FedSSI applies synaptic-importance regularization [30]. GLFC [5] combines class-aware gradient compensation, semantic-relation distillation, and proxy-server guidance to mitigate local and global forgetting. Re-Fed [31] caches replay samples according to local importance and global relevance. FedCIL uses generative replay [32]. DDDR [33] combines pretrained diffusion-based replay with contrastive alignment between real and synthetic data. FOT [34] constructs activation or input subspaces from clients and applies projection to the aggregated server update. CFeD uses knowledge distillation with surrogate data [35]. TARGET [36] distills old-task knowledge from the previous global model using generator-produced samples. FedSubMerge instead merges gradient subspaces at the server and constrains local updates without replay.

III. Methodology

This work enables replay-free federated continual learning across multiple institutions. It aims to preserve prior knowledge while maintaining plasticity for new tasks. The key challenges are global catastrophic forgetting and excessive constraints under client heterogeneity. FedSubMerge addresses the first challenge by merging client PGSs into a shared PGS. FedSubMerge-AD addresses the second through layer-wise peer selection and personalized merging based on subspace similarity. During later tasks, clients project local gradients away from the returned PGS to reduce interference with prior knowledge. Fig. 1 summarizes client PGS construction, server-side PGS merging, and projected local training. The

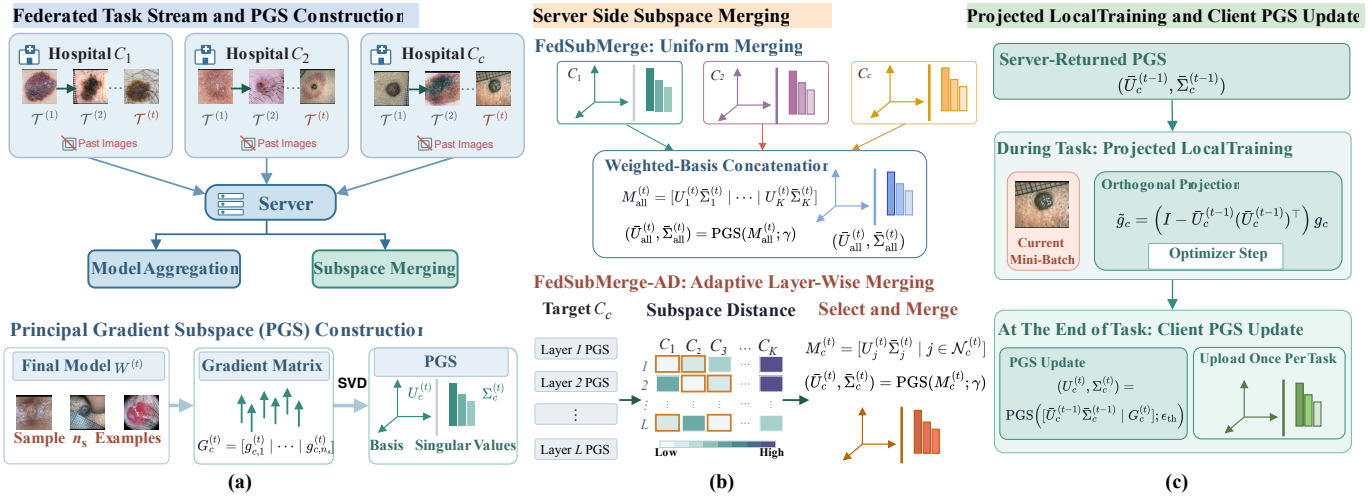


Fig. 1. Overview of the proposed method. (a) Clients perform round-wise federated training and construct layer-wise principal gradient subspaces (PGSs) without replay. (b) The server merges PGSs from all clients (FedSubMerge) or each client's PGS with those of similar peers at each layer (FedSubMerge-AD). (c) Clients project new-task gradients onto the orthogonal complements of the returned PGSs and update their PGSs at each task boundary.

theoretical rationale for these two design choices is presented in Section IV.

A. Problem Formulation

We consider K medical institutions $\mathcal{C} = \{C_1, \ldots, C_K\}$ coordinated by a central server. The system learns a sequence of T tasks $\{\mathcal{T}^{(1)}, \ldots, \mathcal{T}^{(T)}\}$ that arrive over time. For task $\mathcal{T}^{(t)}$, client C_c owns a private dataset

$$\mathcal{D}_c^{(t)} = \{(x_{c,i}^{(t)}, y_{c,i}^{(t)})\}_{i=1}^{N_c^{(t)}}, \tag{1}$$

where $x_{c,i}^{(t)}$ is a medical image and $y_{c,i}^{(t)}$ is its diagnostic label. The dataset $\mathcal{D}_c^{(t)}$ never leaves the institution. During task t, only the current dataset $\mathcal{D}_c^{(t)}$ is available. Images from earlier tasks cannot be accessed or replayed. In many medical settings, institutional consent, storage, and data-use rules may prevent access to images from earlier tasks.

We follow the task-incremental protocol: the task identity is known at test time and selects the matching classifier head. PGS projection is applied to the shared feature extractor, while the task-specific heads are updated normally. We use W for the parameters optimized by FL and omit the index of the task-specific classifier head in f_W for clarity. In our experiments, PGSs are built for trainable convolutional and fully connected weight layers.

Each task is trained for R communication rounds. Let $W^{(t,r)}$ denote the server model after round r of task t. We set $W^{(t,0)} = W^{(t-1)}$ for $t > 1$ and $W^{(t)} \triangleq W^{(t,R)}$. At round r, each client starts from $W^{(t,r-1)}$ and performs E local epochs on $\mathcal{D}_c^{(t)}$. It then uploads the updated parameters $W_c^{(t,r)}$. The server aggregates by

$$W^{(t,r)} = \sum_{c=1}^{K} \alpha_c^{(t)} W_c^{(t,r)}, \qquad \alpha_c^{(t)} = \frac{N_c^{(t)}}{\sum_{j=1}^{K} N_j^{(t)}}, \tag{2}$$

where uniform client weights can also be used when all institutions are treated equally. Equation (2) assumes that every client has current-task data. If a client has no data for task t, it is excluded from that task's model aggregation and PGS upload. Its local PGS remains unchanged.

After observing t tasks, the ideal FCL objective is

$$\min_{W} \sum_{\tau=1}^{t} \sum_{c=1}^{K} \alpha_c^{(\tau)} \, \mathbb{E}_{(x,y)\sim\mathcal{D}_c^{(\tau)}} [\ell(f_W(x), y)], \tag{3}$$

subject to the replay-free constraint that only current-task data are available. The practical goal is therefore to learn the current task while mitigating interference with tasks learned before.

B. Principal Gradient Subspace Construction

FedSubMerge represents earlier tasks with a compact set of important gradient directions. A new update can cause forgetting when it increases the loss on a previous task. For a previous task $\tau < t$ and a small parameter update ΔW, a first-order expansion gives

$$\mathcal{L}^{(\tau)}(W + \Delta W) - \mathcal{L}^{(\tau)}(W) \approx \langle \nabla_W \mathcal{L}^{(\tau)}(W), \Delta W \rangle. \tag{4}$$

Thus, future updates have a small first-order effect on the loss of a previous task when they avoid its main gradient directions. FedSubMerge constructs a shared PGS for each trainable convolutional or fully connected weight layer in the shared feature extractor and returns the resulting layer-wise PGSs to all clients.

After the final aggregation of task t, clients use the final model $W^{(t)}$ to update their PGSs $(U_c^{(t)}, \Sigma_c^{(t)})$. They upload the updated PGSs to the server. The server merges them and returns $(\bar{U}_c^{(t)}, \bar{\Sigma}_c^{(t)})$ for task $t+1$. For an orthonormal basis U, let $P_U = UU^\top$ denote projection onto its span and $P_U^\perp = I - UU^\top$ projection onto the orthogonal complement. If U is empty, we set $P_U = 0$.

To build a PGS, C_c samples n_s training examples from $\mathcal{D}_c^{(t)}$. For a layer of vectorized dimension d, it forms the example-wise gradient matrix

$$G_c^{(t)} = \left[g_{c,1}^{(t)} \,\middle|\, \cdots \,\middle|\, g_{c,n_s}^{(t)} \right] \in \mathbb{R}^{d \times n_s}, \tag{5}$$

where each column is the vectorized per-sample gradient

$$g_{c,i}^{(t)} = \mathrm{vec}\Big(\nabla_W \ell(f_{W^{(t)}}(x_{c,i}^{(t)}), y_{c,i}^{(t)})\Big). \tag{6}$$

Let $G = \hat{U}\hat{\Sigma}\hat{V}^\top$ be the thin SVD, with singular values $\hat{\sigma}_1 \geq \hat{\sigma}_2 \geq \cdots \geq 0$. For an energy threshold $\rho \in (0, 1]$, we keep the smallest number of directions that capture a ρ fraction of the total squared singular values:

$$k(G, \rho) = \min k \quad \text{s.t.} \quad \frac{\sum_{i=1}^{k} \hat{\sigma}_i^2}{\sum_{i \geq 1} \hat{\sigma}_i^2} \geq \rho. \tag{7}$$

The PGS extraction operator is

$$\mathrm{PGS}(G; \rho) \triangleq (U_G, \Sigma_G), \tag{8}$$

where U_G contains the first $k(G, \rho)$ columns of $\hat{U}$, and Σ_G contains the corresponding singular values. If $G = 0$, then $\mathrm{PGS}(G; \rho) = (\emptyset, \emptyset)$. We use ϵ_{th} for client PGS updates and γ for server merging. The singular values preserve the relative strength of the selected gradient directions.

Clients send low-rank bases and singular values at task boundaries. Raw data and replay buffers are not transmitted. Because PGS communication occurs once per task, standard model updates remain the main communication cost.

C. FedSubMerge: Uniform Server-side Subspace Merging

To aggregate previous-task gradient directions learned across institutions, FedSubMerge constructs one shared PGS for each trainable convolutional or fully connected weight layer in the shared feature extractor and returns it to all clients. After task t, the server collects the most recent PGS $(U_c^{(t)}, \Sigma_c^{(t)})$ of every client. It uses the newly uploaded PGSs for participating clients and retains the latest stored PGSs for the others. The server weights each basis by its singular values and concatenates these weighted bases column-wise into a single merge matrix:

$$M_{\mathrm{all}}^{(t)} = \big[\, U_1^{(t)}\Sigma_1^{(t)} \;\big|\; \cdots \;\big|\; U_K^{(t)}\Sigma_K^{(t)} \,\big]. \tag{9}$$

It then applies the PGS operator to the concatenated matrix to extract a compact shared PGS:

$$(\bar{U}_{\mathrm{all}}^{(t)}, \bar{\Sigma}_{\mathrm{all}}^{(t)}) = \mathrm{PGS}\Big(M_{\mathrm{all}}^{(t)}; \gamma\Big), \tag{10}$$

where $\gamma \in (0, 1]$ is the server-side energy threshold. This shared PGS is sent to all clients:

$$(\bar{U}_c^{(t)}, \bar{\Sigma}_c^{(t)}) = (\bar{U}_{\mathrm{all}}^{(t)}, \bar{\Sigma}_{\mathrm{all}}^{(t)}), \qquad c = 1, \ldots, K. \tag{11}$$

Multiplying each basis by its singular values preserves the strength of its gradient directions. The SVD also combines shared directions across clients. A larger γ preserves more gradient energy and more directions from previous tasks, while a smaller value produces a lower-rank PGS. The SVD recompression also limits rank growth as tasks accumulate.

A shared PGS may contain directions that are not useful for every client, especially when client data differ strongly. FedSubMerge-AD reduces this restriction by returning a personalized PGS.

D. FedSubMerge-AD: Adaptive Layer-wise Subspace Merging

To reduce constraints from less related clients while retaining useful cross-client knowledge, FedSubMerge-AD constructs a personalized PGS for each client at each selected layer. Starting from the most recent client PGSs collected by the server, it compares the target client's PGS with those of the other clients and selects the n most similar peers independently at each layer. Layer-wise selection is used because PGS similarity can vary across the network.

For clients C_c and $C_{c'}$, define $k_c^{(t)} = \mathrm{rank}(U_c^{(t)})$. We also set $r_{c,c'}^{(t)} = \min(k_c^{(t)}, k_{c'}^{(t)})$. The principal angles between their PGSs follow from the singular values of the cross-correlation matrix:

$$\cos\theta_{c,c',i}^{(t)} = \sigma_i\Big((U_c^{(t)})^\top U_{c'}^{(t)}\Big), \qquad i = 1, \ldots, r_{c,c'}^{(t)}. \tag{12}$$

We then use the following projection distance, which is well-defined for subspaces with different ranks:

$$D_{c,c'}^{(t)} = \sqrt{\sum_{i=1}^{r_{c,c'}^{(t)}} \sin^2\theta_{c,c',i}^{(t)} + \frac{1}{2}\left|k_c^{(t)} - k_{c'}^{(t)}\right|}. \tag{13}$$

Equivalently,

$$D_{c,c'}^{(t)} = \frac{1}{\sqrt{2}}\left\|P_{U_c^{(t)}} - P_{U_{c'}^{(t)}}\right\|_F. \tag{14}$$

A smaller $D_{c,c'}^{(t)}$ means that the two client PGSs contain more similar gradient directions. The rank term allows PGSs with different numbers of directions to be compared.

For target client C_c, the server selects the n nearest peers:

$$\mathcal{P}_c^{(t)} = \underset{\substack{\mathcal{P} \subseteq \{1,\ldots,K\}\setminus\{c\} \\ |\mathcal{P}| = n}}{\arg\min} \sum_{c' \in \mathcal{P}} D_{c,c'}^{(t)}, \qquad \mathcal{N}_c^{(t)} = \{c\} \cup \mathcal{P}_c^{(t)}. \tag{15}$$

For the resulting layer-specific peer set, the server weights the PGS basis of the target client and each selected peer by its singular values, then concatenates these weighted bases column-wise:

$$M_c^{(t)} = \big[\, U_j^{(t)}\Sigma_j^{(t)} \;\big|\; j \in \mathcal{N}_c^{(t)} \,\big], \tag{16}$$

Applying the same PGS operator to this concatenated matrix gives the personalized PGS:

$$(\bar{U}_c^{(t)}, \bar{\Sigma}_c^{(t)}) = \mathrm{PGS}\Big(M_c^{(t)}; \gamma\Big). \tag{17}$$

Thus, FedSubMerge-AD uses the same concatenation-and-recompression procedure as FedSubMerge, but includes only the target client and its selected peers at each layer. The neighborhood size n controls how much information is shared. When $n = K-1$, FedSubMerge-AD reduces to FedSubMerge. When $n = 0$, only the target client's PGS is used. Intermediate values balance the use of cross-client previous-task directions and client-specific plasticity (Section V-D).

Algorithm 1 FedSubMerge and FedSubMerge-AD

Require: Clients $\{C_c\}_{c=1}^{K}$, tasks $\{\mathcal{T}^{(t)}\}_{t=1}^{T}$, rounds R, local epochs E, γ, ϵ_{th}, and n
Ensure: Final model $W^{(T)}$ and returned PGSs

```
Initialize W^(0) and all PGSs as (∅, ∅)
for t = 1, ..., T do
    W^(t,0) ← W^(t−1)
    for r = 1, ..., R do
        Clients train for E epochs and upload W_c^(t,r); use Eq. (18) if t ≥ 2
        Server computes W^(t,r) by Eq. (2)
    end for
    Set W^(t) ← W^(t,R) and send it to the clients
    Clients form G_c^(t), update their PGSs by Eq. (19), and upload the updated PGSs
    if n = K − 1 then                                   ▷ FedSubMerge
        Return the shared PGS by Eqs. (9)– (11)
    else                                                ▷ FedSubMerge-AD
        Select peers by Eqs. (13)– (15)
        Return personalized PGSs by Eqs. (16)– (17)
    end if
end for
return W^(T) and the returned PGSs
```

E. Projected Local Training and Client PGS Update

Only gradients in the onvolutional and fully connected weight layer in the shared feature extractor are projected. Clients train normally on the first task because no earlier PGS is available. For $t \geq 2$, client C_c receives $(\bar{U}_c^{(t-1)}, \bar{\Sigma}_c^{(t-1)})$. For each selected layer and mini-batch gradient g_c, it computes

$$\tilde{g}_c = P^{\perp}_{\bar{U}_c^{(t-1)}} g_c = g_c - \bar{U}_c^{(t-1)} \big(\bar{U}_c^{(t-1)}\big)^{\top} g_c. \tag{18}$$

The projected gradient replaces the original gradient before the optimizer step. Other parameters, including task-specific classifier heads, are updated normally. For FedSubMerge, all clients use the same returned PGS.

At the end of task t, client C_c updates its PGS by combining the PGS received from the server after task $t-1$ with the current-task gradient matrix:

$$(U_c^{(t)}, \Sigma_c^{(t)}) = \text{PGS}\Big(\Big[\bar{U}_c^{(t-1)}\bar{\Sigma}_c^{(t-1)} \;\Big|\; G_c^{(t)}\Big]; \epsilon_{\text{th}}\Big), \tag{19}$$

with $(\bar{U}_c^{(0)}, \bar{\Sigma}_c^{(0)}) = (\emptyset, \emptyset)$. For FedSubMerge, $(\bar{U}_c^{(t-1)}, \bar{\Sigma}_c^{(t-1)})$ is the shared PGS returned to all clients. For FedSubMerge-AD, it is the personalized PGS returned to client C_c. The updated $(U_c^{(t)}, \Sigma_c^{(t)})$ is then uploaded to the server for merging.

IV. Theoretical Analysis

This section explains how PGS merging mitigates global forgetting and how adaptive merging preserves plasticity for the incoming task.

A. Why Can PGS Merging Mitigate Global Forgetting?

Let $U = \bar{U}_{\text{all}}^{(t-1)}$ be the shared PGS and $P_U^{\perp} = I - UU^{\top}$. If client C_c applies $\Delta w_c = P_U^{\perp} z_c$, the FedAvg update becomes

$$\Delta w = \sum_c \alpha_c \Delta w_c = P_U^{\perp} \sum_c \alpha_c z_c, \qquad U^{\top}\Delta w = 0. \tag{20}$$

Thus, aggregation does not reintroduce update components along the historical directions protected by U.

For a previous task $\tau < t$, let $\mathcal{L}_{\text{g}}^{(\tau)}(w)$ denote its global loss at model w and $h_\tau(w) = \nabla \mathcal{L}_{\text{g}}^{(\tau)}(w)$. Define the uncovered gradient component as $\varepsilon_\tau = \|P_U^{\perp} h_\tau(w)\|$. If $\mathcal{L}_{\text{g}}^{(\tau)}$ is L_τ-smooth, then

$$\begin{aligned} &\mathcal{L}_{\text{g}}^{(\tau)}(w + \Delta w) - \mathcal{L}_{\text{g}}^{(\tau)}(w) \\ &\leq \underbrace{\varepsilon_\tau \|\Delta w\|}_{\text{first-order interference}} + \underbrace{\frac{L_\tau}{2}\|\Delta w\|^2}_{\text{second-order term}}. \end{aligned} \tag{21}$$

The first term is determined by previous task directions not covered by the merged PGS and vanishes when $h_\tau(w) \in \text{span}(U)$. By combining historical directions from multiple clients, FedSubMerge can reduce ε_τ and protect global historical knowledge.

B. Why Can Adaptive Merging Preserve Plasticity?

For the current-task gradient g_c, define the retained gradient energy as $\mathcal{R}(U; g_c) = \|P_U^{\perp} g_c\|^2$. Let $\mathcal{S}_{\text{all}} = \text{span}(U^{\text{all}})$ and $\mathcal{S}_c^{\text{AD}} = \text{span}(U_c^{\text{AD}})$. Without SVD truncation, adaptive merging uses a subset of the client subspaces used by uniform merging, so $\mathcal{S}_c^{\text{AD}} \subseteq \mathcal{S}_{\text{all}}$. Therefore,

$$\begin{aligned} &\mathcal{R}(U_c^{\text{AD}}; g_c) - \mathcal{R}(U^{\text{all}}; g_c) \\ &\quad = \big\|P_{\mathcal{S}_{\text{all}} \ominus \mathcal{S}_c^{\text{AD}}} g_c\big\|^2 \geq 0. \end{aligned} \tag{22}$$

This term measures the gradient energy for the current task that is removed by uniform merging but retained by adaptive merging. Thus, adaptive merging avoids unnecessary constraints and preserves at least as much gradient energy for learning the incoming task. If the selected subspace already covers the relevant historical directions, this additional plasticity does not weaken first-order protection. With SVD truncation, the exact inclusion may not hold, and n and γ jointly control the empirical stability–plasticity trade-off.

V. Experiments

We evaluated FedSubMerge on four medical-image benchmarks: PathMNIST, OrganAMNIST, Hyper-Kvasir, and a real-world multi-center skin-lesion benchmark constructed from four source datasets. Together, they cover distribution-based label skew, quantity-based label skew, and real-world multi-center feature skew. We compared final ACC and BWT with representative FL, CL, and FCL methods. We also analyzed robustness, design ablations, and communication overhead.

TABLE I
EXPERIMENTAL SETUPS. ONLY THE TRAINING DATA WERE PARTITIONED ACROSS CLIENTS.

Setting	Scenario	Dataset	#Classes	Federated split	Task split	#Clients	#Samples
Label skew	Distribution-based	PathMNIST	9	Dirichlet ($\alpha \in \{0.1, 0.3\}$)	$[2, 2, 2, 3]$	10	107180
		Hyper-Kvasir	20	Dirichlet ($\alpha \in \{0.1, 0.3\}$)	$[2] \times 10$	10	10662
	Quantity-based	OrganAMNIST	11	Random 2 classes subsets per client	$[3, 4, 4]$	10	58830
		Hyper-Kvasir	20	Random 50%-class subsets per client	$[4, 4, 4, 4, 4]$	10	10662
Feature skew	Real-world multi-center skin lesion	HAM10000	6	One local dataset per client	$[2, 2, 2]$	4	9041
		Derm7pt					1926
		BCN20000					12148
		DMF					1065

A. Experimental Setup

Datasets and Task Streams: Table I summarizes the datasets and task streams. We considered three forms of non-IID heterogeneity. Under *distribution-based label skew* (distribution skew), clients share a label space but have different local class proportions. We simulated this setting with Dirichlet partitioning. Under *quantity-based label skew* (quantity skew), clients observe different class subsets, instantiated with disjoint or uneven subsets or random 50%-class subsets. Under *feature skew*, the image distributions vary across centers because of site-specific acquisition or population characteristics, while the label space is shared. Accordingly, each source dataset in the multi-center skin-lesion benchmark was treated as one client.

PathMNIST [37] comprises nine colorectal histopathology classes and used Dirichlet partitioning with $\alpha \in \{0.1, 0.3\}$ [38]. **OrganAMNIST** [37] comprises 11 abdominal CT organ classes distributed as random 2 classes subsets per client for each task. For **Hyper-Kvasir** [39], we retained the 20 largest classes and evaluated both distribution and quantity skew. The **real-world multi-center skin lesion** setting combined HAM10000 [40], Derm7pt [41], BCN20000 [42], and DMF [43]. The four clients shared six classes.

For PathMNIST and OrganAMNIST, we used the official training sets for model training, the official validation sets for hyperparameter selection, and the official test sets for global testing. Hyper-Kvasir was split into 70% training, 10% global validation, and 20% global test data. Each skin-lesion center was split locally using the same $70/10/20\%$ proportions, and the local validation and test subsets were combined into global validation and test sets, respectively. Only the training data were partitioned across clients as specified in Table I.

Evaluation Protocol and Metrics: For a fair comparison, all methods used the same task order, data splits, ResNet-18 backbone [44], and evaluation schedule. Task identity selected the classifier head, and after each task the global model was evaluated on all tasks seen so far. Hyperparameters were not tuned separately for individual clients or tasks, and the global test sets were reserved for final reporting.

We report average accuracy (ACC) and backward transfer (BWT) [45]. Let $a_{t,\tau}$ be the test accuracy on task τ after training through task t. After the final task T,

$$\mathrm{ACC} = \frac{1}{T}\sum_{\tau=1}^{T} a_{T,\tau}, \qquad \mathrm{BWT} = \frac{1}{T-1}\sum_{\tau=1}^{T-1}\left(a_{T,\tau} - a_{\tau,\tau}\right). \tag{23}$$

Higher ACC and higher (i.e., less negative) BWT indicate better final performance and less degradation on earlier tasks, respectively.

Implementation Details: All models were initialized from scratch and optimized with SGD using a batch size of 64. PathMNIST and OrganAMNIST used their original 128×128 resolution from MedMNIST v2 dataset, whereas Hyper-Kvasir and skin-lesion images were resized to 256×256. Each task comprised $R = 20$ communication rounds with $E = 3$ local epochs per round. The initial learning rate was 10^{-2} and was reduced tenfold at 50% and 75% of training. At each task boundary, every client sampled $n_s = 256$ current-task inputs to update its PGS. We set $\gamma = 0.97$ for server-side merging and $\epsilon_{\mathrm{th}} = 0.99$ for client PGS updates. For FedSubMerge-AD, the number of selected peers was chosen once per setting on the global validation set: $n = 4$ for PathMNIST, $n = 5$ for Hyper-Kvasir with distribution skew, $n = 3$ for OrganAMNIST, $n = 6$ for Hyper-Kvasir with quantity skew, and $n = 2$ for the multi-center skin-lesion setting. Section V-D examines sensitivity to n and γ. All experiments used PyTorch 1.13 on one NVIDIA A100 GPU.

Competing Methods: Table II compares the proposed methods with generic FL, CL methods adapted to FL, and dedicated FCL methods. The FL methods were FedAvg [1], FedProx [7], and SCAFFOLD [8]. Among the CL methods, Fed-EWC [20], Fed-SI [21], and Fed-GPM [6] were replay-free, whereas Fed-DER [18] was replay-based. The FCL methods included the replay-free method FOT [34] and the replay-based methods GLFC [5], Re-Fed [31], TARGET [36], and DDDR [33]. GLFC and Re-Fed stored samples from previous tasks in a buffer, while TARGET and DDDR generated samples for previous tasks. To enable a common evaluation, GLFC, TARGET, DDDR, and Re-Fed were adapted to task-specific classifier heads at inference. Replay-based methods are marked with an asterisk in Table II. For Fed-DER, GLFC, and Re-Fed, the buffer size was fixed at 256 samples. All replay-free methods, including ours, trained only on samples from the current task.

TABLE II

MAIN ACC AND BWT RESULTS (%, HIGHER IS BETTER). BOLD AND UNDERLINED MARK THE FIRST- AND SECOND-RANKED VALUES. * MARKS REPLAY-BASED METHODS. FED-DER, GLFC, AND RE-FED USE STORED SAMPLES, WHILE TARGET AND DDDR GENERATE SAMPLES FOR PREVIOUS TASKS.

	PathMNIST				**Hyper-Kvasir**				**OrganAMNIST**		**Hyper-Kvasir**		**Skin Lesion**	
	$\alpha = 0.3$		$\alpha = 0.1$		$\alpha = 0.3$		$\alpha = 0.1$		Quantity skew		Quantity skew		Feature skew	
Method	ACC	BWT	ACC	BWT	ACC	BWT	ACC	BWT	ACC	BWT	ACC	BWT	ACC	BWT
FL methods														
FedAvg [1]	67.23	-25.70	57.35	-35.18	47.58	-29.27	42.45	-33.13	40.20	-45.74	48.15	-35.66	64.45	-20.63
FedProx [7]	68.51	-20.12	58.45	-33.88	63.44	-17.58	55.87	-28.89	41.33	-43.39	46.40	-38.40	67.90	-17.45
SCAFFOLD [8]	73.48	-21.86	59.95	-32.55	63.82	-16.30	62.08	-20.19	46.29	-36.21	51.14	-30.84	72.33	-14.82
CL methods adapted to FL														
Fed-EWC [20]	76.26	-17.73	64.38	-26.40	64.46	-17.61	75.03	-11.83	52.59	-31.37	53.15	-28.71	73.18	-14.65
Fed-SI [21]	77.22	-18.87	65.17	-27.20	63.58	-18.13	71.89	-15.63	48.35	-32.23	45.08	-30.54	77.72	-15.13
Fed-GPM [6]	81.40	-13.70	69.06	-19.80	66.76	-15.96	64.22	-16.63	61.20	-22.45	61.52	-19.21	82.76	-8.41
Fed-DER* [18]	81.70	-13.03	72.85	-18.45	69.64	-14.35	66.01	-19.34	62.57	-18.66	61.18	-18.85	78.86	-12.35
FCL methods														
GLFC* [5]	84.05	-13.06	75.30	-17.80	72.16	-12.92	66.59	-21.60	64.10	-17.50	63.06	-15.95	79.45	-7.80
TARGET* [36]	84.75	-7.45	76.40	-15.85	71.02	-14.08	67.10	-18.21	65.20	-16.80	67.24	-8.21	80.15	-5.40
FOT [34]	86.20	-6.30	74.85	-16.40	70.03	-11.30	74.20	-15.47	66.85	-15.20	66.57	-4.53	80.80	-7.65
Re-Fed* [31]	87.13	-5.31	**80.72**	-15.41	73.96	-13.11	73.79	-12.88	68.20	-13.85	69.38	-1.24	84.21	-6.20
DDDR* [33]	88.93	-4.01	77.30	-14.87	75.27	-12.15	**77.24**	**-10.30**	69.35	**-11.85**	68.94	-2.29	84.43	-5.95
Ours														
FedSubMerge	90.97	**-0.86**	77.50	**-14.30**	**78.64**	**-6.55**	76.64	-10.55	70.17	-14.66	70.76	**0.07**	85.53	**-4.52**
FedSubMerge-AD	**93.42**	-3.37	79.40	-16.85	78.12	-10.95	76.16	-13.05	**74.42**	-15.92	**70.97**	-1.47	**85.75**	-5.20

B. Overall Performance

Table II reports ACC and BWT on four medical image classification benchmarks across seven federated settings. For ACC, one of the two proposed variants ranked first among replay-free methods in every setting. When replay-based methods were also included, the proposed variants ranked first in five settings and second on PathMNIST and Hyper-Kvasir at $\alpha = 0.1$. For example, FedSubMerge-AD reached 93.42% ACC on PathMNIST at $\alpha = 0.3$, compared with 88.93% for DDDR. Under quantity skew on Hyper-Kvasir, it also outperformed Re-Fed, achieving 70.97% versus 69.38%. The proposed variants additionally ranked first on Hyper-Kvasir at $\alpha = 0.3$, OrganAMNIST, and the real-world multi-center skin-lesion benchmark.

The advantage of the proposed methods over other replay-free methods was consistent across imaging modalities and heterogeneity types. Compared with the strongest replay-free baseline in each setting, the proposed variants improved ACC by 1.61–8.61 percentage points. The largest difference was observed on Hyper-Kvasir at $\alpha = 0.3$, where FedSubMerge reached 78.64%, compared with 70.03% for FOT. On OrganAMNIST, FedSubMerge-AD reached 74.42%, compared with 66.85% for FOT. Under the more severe distribution skew on Hyper-Kvasir, FedSubMerge reached 76.64%, compared with 75.03% for Fed-EWC. FedSubMerge-AD also improved the real-world multi-center skin-lesion result from 82.76% for Fed-GPM to 85.75%.

FedSubMerge and FedSubMerge-AD showed complementary strengths in stability and plasticity. FedSubMerge-AD achieved higher ACC in five of the seven settings, whereas FedSubMerge had higher BWT than FedSubMerge-AD in every setting and ranked first in BWT in five. In particular, FedSubMerge obtained BWT values close to zero on PathMNIST at $\alpha = 0.3$ (-0.86%) and Hyper-Kvasir under quantity skew (0.07%), indicating that earlier-task performance was largely retained. DDDR ranked first in BWT in the remaining two settings. On OrganAMNIST, its higher BWT but lower final ACC than FedSubMerge-AD further illustrates that BWT and ACC characterize different aspects of continual-learning performance. Overall, the results are consistent with the intended stability–plasticity trade-off: uniform merging emphasizes the preservation of earlier tasks, whereas adaptive merging retains greater capacity for learning incoming tasks under client heterogeneity.

C. Robustness to Heterogeneity and Client Participation

We examined robustness to label skew and reduced client participation under different numbers of clients.

1) Effect of Data Heterogeneity: Table III(a) shows that all methods were affected by stronger label skew. For example, at $\alpha = 0.05$, FedSubMerge-AD achieved 75.20% ACC, compared with 54.65% for FedAvg. Across all heterogeneity

TABLE III
ROBUSTNESS TO DIRICHLET LABEL SKEW, CLIENT SCALING, AND PARTIAL PARTICIPATION.

(a) Robustness to Dirichlet label skew on PathMNIST.

	$\alpha = 0.05$		$\alpha = 0.1$		$\alpha = 0.3$		$\alpha = 0.5$		IID	
Method	ACC	BWT	ACC	BWT	ACC	BWT	ACC	BWT	ACC	BWT
FedAvg	54.65	-41.20	57.35	-35.18	67.23	-25.70	75.05	-19.45	82.10	-13.05
GLFC	67.40	-25.05	75.30	-17.80	84.05	-13.06	85.65	-10.25	87.30	-8.10
FedSubMerge	73.10	**-19.30**	77.50	**-14.30**	90.97	**-0.86**	91.10	**-1.05**	91.95	**-0.62**
FedSubMerge-AD	**75.20**	-21.95	**79.40**	-16.85	**93.42**	-3.37	**93.65**	-2.45	**93.95**	-1.80

(b) Scalability of FedSubMerge-AD across client populations and participation rates on PathMNIST ($\alpha = 0.3$, $n = 4$). Overhead is the total PGS communication volume relative to model communication over 20 FedAvg rounds.

	$K = 10$			$K = 20$			$K = 40$		
Participation Rate	ACC	BWT	Overhead	ACC	BWT	Overhead	ACC	BWT	Overhead
1.00	93.42	-3.37	3.0%	92.65	-4.20	3.1%	90.20	-6.95	3.5%
0.50	91.55	-5.80	5.9%	90.40	-7.05	6.1%	88.10	-9.05	6.4%
0.30	88.20	-8.25	11.8%	87.42	-8.17	12.0%	84.85	-12.40	12.3%

levels, FedSubMerge-AD gave the highest ACC, whereas FedSubMerge consistently achieved the best BWT. Overall, adaptive merging favored ACC, while uniform merging provided greater stability under label skew.

2) Effect of Client Participation: Table III(b) evaluates FedSubMerge-AD at different participation rates for $K = 10$, 20, and 40. Reducing client participation affected both ACC and BWT across all client settings. For example, with $K = 40$, decreasing the participation rate from 1.00 to 0.30 reduced ACC from 90.20% to 84.85% and changed BWT from -6.95% to -12.40%. This indicates that fewer participating clients weakened both overall performance and knowledge preservation. FedSubMerge-AD introduced modest additional communication overhead. Under full participation, the PGS exchange accounted for 3.0%–3.5% of model communication across the three client settings. Its relative share increased as participation decreased because model communication involved fewer clients per round, whereas PGSs were communicated once per task.

D. Ablation Studies

We analyzed the components on PathMNIST with $\alpha = 0.3$, the task-wise behavior on Hyper-Kvasir with $\alpha = 0.3$, and hyperparameter sensitivity on the real-world multi-center skin-lesion setting.

1) PGS Projection and Cross-Client Merging: Table IV compares FedAvg, PGS projection using only the target client's subspace, and server-side PGS merging. Local PGS projection already reduces forgetting, but cross-client merging provides a further gain, indicating that a client cannot preserve knowledge that exists only in other institutions using its own PGS alone. Adaptive selection achieves the highest ACC, whereas uniform merging across all clients gives the higher BWT. The contrast is consistent with their intended roles: selective merging preserves greater plasticity, whereas a broader shared subspace places greater emphasis on stability across earlier tasks.

2) PGS Source and Merging Design: Gradient PGSs perform better than activation PGSs because they directly encode the parameter directions that affect the training loss. Similarity-based peer selection also improves ACC over random selection, while their BWT values are close. Peer relevance therefore contributes mainly to learning the incoming tasks. Using singular-value weights improves both metrics relative to the unweighted variant, showing that the server should preserve the relative importance of merged directions rather than treat them uniformly.

3) Hyperparameter Sensitivity: Fig. 2 shows how the neighborhood size and energy threshold affect ACC and BWT on the real-world multi-center setting. Selecting two peers gives the best balance. Using the target client's PGS alone omits useful gradient information from other institutions, whereas merging all peers slightly improves BWT at the cost of ACC. Varying γ shows a similar pattern. The highest ACC is obtained at $\gamma = 0.97$, while increasing the energy threshold improves BWT but places a stronger constraint on the update. The selected configuration therefore avoids both insufficient sharing and excessive constraints.

4) Task-wise Stability–Plasticity Analysis: The accuracy matrices in Fig. 3 explain the task-wise results on Hyper-Kvasir at $\alpha = 0.3$. FedSubMerge-AD generally has higher diagonal entries, indicating stronger learning ablility, but its old task accuracies decline more sharply. Thus, adaptive selection favors plasticity, whereas uniform merging provides greater stability.

VI. DISCUSSION

The results indicate that gradient-subspace sharing provides a useful mechanism for replay-free continual medical image

TABLE IV

ABLATION OF DESIGN VARIANTS ON PATHMNIST ($\alpha = 0.3$). Δ DENOTES THE SIGNED DIFFERENCE RELATIVE TO THE FEDSUBMERGE-AD CONFIGURATION WITH $n = 4$.

Configuration	ACC	Δ ACC	BWT	Δ BWT
No PGS projection (FedAvg)	67.23	−26.19	−25.70	−22.33
Local PGS projection ($n = 0$)	82.05	−11.37	−17.60	−14.23
Uniform PGS merging (FedSubMerge, $n = K - 1$)	90.97	−2.45	−0.86	+2.51
Activation PGS	87.80	−5.62	−7.23	−3.86
Random peer selection	88.55	−4.87	−3.24	+0.13
No singular-value weighting	89.60	−3.82	−4.97	−1.60
FedSubMerge-AD ($n = 4$)	93.42	0.00	−3.37	0.00

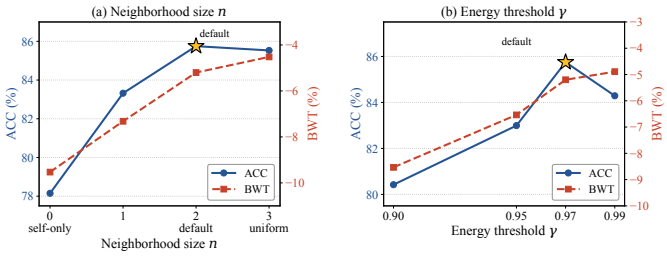


Fig. 2. Hyperparameter sensitivity of FedSubMerge-AD on the real-world multi-center skin-lesion feature-skew setting. (a) Neighborhood size n. With $K = 4$, $n = 0$ uses only the target client's PGS and $n = 3$ uses all available peers, corresponding to uniform merging across all clients. (b) Server-side energy threshold. Stars mark the default configuration, $n = 2$ and $\gamma = 0.97$.

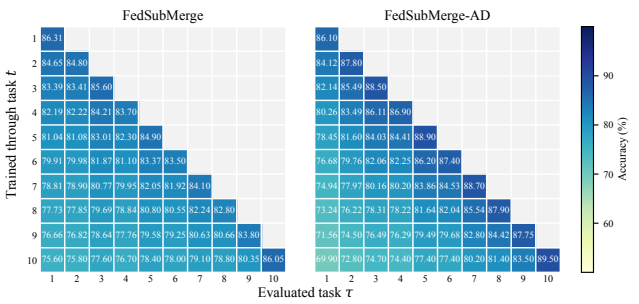


Fig. 3. Task-wise accuracy matrices of FedSubMerge and FedSubMerge-AD on Hyper-Kvasir ($\alpha = 0.3$). Each cell gives the accuracy on task τ after training through task t. Diagonal entries reflect new task learning, while their changes to the corresponding final row entries reflect forgetting.

classification. A local PGS represents the previous tasks encountered at one institution, whereas server-side merging also incorporates directions learned at other sites without transmitting or replaying earlier patient images. This is relevant when rare conditions, new diagnostic targets, or acquisition changes are distributed across hospitals and historical data cannot be revisited.

Uniform and adaptive merging show different stability–plasticity trade-offs. Uniform merging covers a broader set of previous-task directions and generally retains earlier tasks better. FedSubMerge-AD excludes less related layer-wise directions and preserves more flexibility for centers with different scanners, populations, or task histories. Results across histopathology, abdominal CT, gastrointestinal endoscopy, and real-world multi-center dermoscopy indicate that this trade-off appears across modalities. The flexibility of FedSubMerge-AD may be suitable for FCL with long task streams. Maintaining sufficient optimization freedom becomes increasingly important as institutions continually encounter new diseases, imaging protocols, and patient populations. FedSubMerge-AD therefore provides a promising option for sustained adaptation under continuously evolving clinical demands.

The PGS in our method is constructed from gradients and is not itself a privacy guarantee. Deployment would require appropriate secure aggregation, encryption, or privacy-preserving perturbation [46].

VII. CONCLUSION

In this work, we propose FedSubMerge, a replay-free framework for continual medical image classification across medical institutions. FedSubMerge merges compact client gradient subspaces into a shared protective subspace, thereby preserving diagnostic knowledge distributed across sites without replaying previous patient images. Building upon this uniform merging framework, we further propose FedSubMerge-AD, which performs layer-wise adaptive selection and personalized subspace merging to preserve greater plasticity under client heterogeneity. Experiments spanning pathology, CT, endoscopy, and dermoscopy support the feasibility of both methods across medical imaging modalities. The present study is limited to synchronous task streams and relatively small backbone models. Future work will investigate asynchronous and longer task streams, larger architectures such as vision transformers, and medical image segmentation tasks.

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